

Both piconets and scatternets are types of networks that are established in an ad hoc fashion. In an ad hoc network, all devices in the network are treated as peers. Even though one device in a piconet will assume the role of master, that role could be assumed by any device, and can be swapped between devices if and when necessary.

Let's compare that to a more structured type of network such as a cellular phone network.

In a typical cellular network, base stations (or transmitters) are placed at strategic positions to provide local cell coverage, while cellular phones (or radios) are used to connect via radio link to the base stations. With a cellular network, there is a strict and permanent separation between the base stations and the terminals. A cellular phone can never function as a base station, and vice versa; only the base station can provide channel access, allocate channels, control network traffic, and otherwise manage the flow of signals through its particular cell.

The advantages of an ad hoc network are many. First, there is no need to establish a costly infrastructure to service a particular geographic region – ad hoc networks are formed on the fly, where the devices are, without the need for base stations to control network communications. Second, unlike cellular networks, multiple ad hoc networks can occupy the same physical space without fear of interference. Third, any Bluetooth unit can “control” an ad hoc piconet as a master device; no special control devices (or operators) are necessary.

**Ad hoc networks
don't need costly
infrastructure, thereby
reducing cost**

So we have now discussed the basics of Bluetooth and how communications are done. There are few issues with Bluetooth connectivity and ad-hoc networks that need to be discussed.

The most important is the question of security. One question that has often been raised is just how secure are these connections? What's to stop someone else's Bluetooth device from picking up your personal signals. For businessmen at crowded locations, or anyone else for that matter that considers their data valuable and needs security, Bluetooth could be a mistake. However, the Bluetooth